

The model is never retrained. Retraining on a subset would test whether those features SUFFICE to build a model — not whether THIS model used them.

Frozen model
 $p(\text{Exploits}) = 0.97$

explanation names top-k features

COMPREHENSIVENESS

mask the top-k → probability should FALL



... k = 10

SUFFICIENCY

keep ONLY the top-k → probability should HOLD



... k = 10

RANDOM CONTROL

mask k RANDOM features — the actual experiment



... k = 10

Curve
+
AUC

per
explainer